

SYNTHETIC FIXTURE ? NOT A REAL INSPECTION RECORD

ULTRASONIC THICKNESS INSPECTION REPORT

Report No. MRPL-SYN-2026-0441
Facility Synthetic Refinery Unit 3 (training data)
Asset Crude overhead line 12-P-204, elbow E-07
Date 2026-08-14
Inspector A. Rao, Level II UT (synthetic persona)

1. SCOPE

Routine ultrasonic thickness survey of elbow E-07 on the crude overhead line, following the 5000-hour interval defined in the site inspection schedule. Nominal wall thickness for this component is 9.5 mm.

2. METHOD

Contact ultrasonic thickness measurement at twelve grid points, using a 5 MHz dual element probe. Surface prepared by wire brushing. Couplant applied at each point. Calibration verified against a step wedge before and after the survey.

3. GENERAL CONDITION

External coating breakdown observed on the outer radius of the elbow over an area of roughly 180 by 120 mm. Product-side scale visible through the breakdown. No through-wall penetration and no active leak was observed at the time of the survey.

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4. MEASURED RESULTS

Grid point	Measured (mm)	Loss from nominal (mm)
G-01	9.4	0.1
G-02	9.1	0.4
G-03	8.8	0.7
G-04	7.9	1.6
G-05	7.1	2.4
G-06	6.8	2.7
G-07	7.4	2.1
G-08	8.2	1.3
G-09	9.0	0.5
G-10	9.3	0.2
G-11	9.4	0.1
G-12	9.2	0.3

Minimum measured thickness	6.8 mm at grid point G-06
Maximum wall loss	2.7 mm at grid point G-06
Nominal wall thickness	9.5 mm

5. OBSERVATIONS

The deepest wall loss is concentrated on the outer radius of the elbow, coincident with the coating breakdown described in section 3. The loss pattern is consistent with localised external corrosion under disbonded coating rather than internal erosion.

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6. PHOTOGRAPH

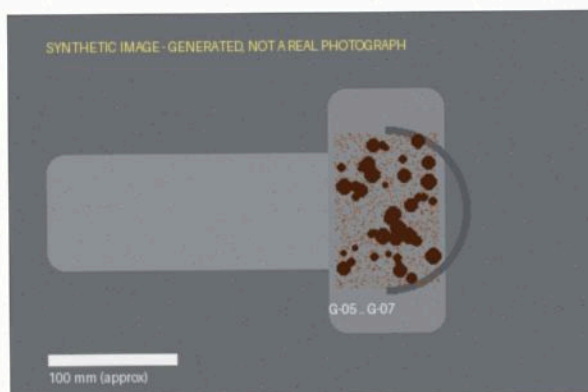
See attached photograph of the affected outer radius, showing coating breakdown and surface product deposits at grid points G-05 through G-07.

7. INSPECTOR NOTE

The measured maximum wall loss of 2.7 mm exceeds the site escalation threshold. This report is referred to the integrity engineer for a fitness-for-service assessment before the unit returns to full rate.

8. NEXT SURVEY

Recommended re-inspection interval reduced to 1000 hours pending the outcome of the fitness-for-service assessment.



check G-06 again after cleaning - AR